
When the reference hides uncertainty: a data-based simulation audit

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Abstract

1 Scoring the same nominal 95% intervals against a sampled reference gives 98.9–
2 99.6% inclusion, versus 92.3–93.6% coverage of the generating-pool effect, in a
3 single-cell simulation audit at 800 cells. An exact variance calculation explains the
4 discrepancy: under a halving of expression, sharing the sampled reference mean
5 removes three quarters of the reference-arm variance contribution from the error
6 being assessed. We trace the audit to a 50-cell reporting rule previously justified
7 by point-estimate recovery. In 5,000 studies per setting, the original interval
8 rejects a true null in 9.8–32.2% of studies; a Welch control without bootstrap
9 resampling noise still rejects in 8.9–30.8%. A further 32,000-study comparison
10 evaluates an interior point of the warning range, larger bootstrap budgets, BCa
11 and bootstrap- t . None of the tested methods satisfies coverage, detection and
12 nominal null calibration jointly at 30 or 50 cells. Rescoring patient- t intervals
13 against fixed targets confirms a coverage/detection tradeoff, without establishing
14 patient-population validity. The evidence is conditional on empirical cell pools: it
15 identifies a failed reporting claim and tests concrete repairs, without validating a
16 replacement threshold or a patient world model.

1 Motivation: evaluate the reported uncertainty

18 A nominal 95% interval can appear to cover its reference almost perfectly while covering the intended
19 parameter less often. In our single-cell simulation audit, changing only the scoring reference changes
20 inclusion from 98.9–99.6% to 92.3–93.6% at 800 cells. The source of the discrepancy is explicit: the
21 estimate and reference share a sampled mean, canceling part of the error that the interval is meant to
22 cover. We derive that variance difference and examine its consequences for a reporting rule.

23 The case comes from our unpublished colorectal project, where point-estimate recovery had been
24 used to justify a 50-cell reporting boundary. We audit that recorded inference using simulations
25 resampled from observed single-cell distributions. The analysis covers GUCA2A, one labeling axis
26 and one weighting, with ten SMC and six KUL3 patients. Each study draws cells from a two-patient
27 pool; repeated draws quantify performance conditional on those source pools, not uncertainty over
28 new patients. The contribution is this documented failure, its quantitative explanation and paired
29 interval comparisons.

30 Section 2 assesses the original interval, a resampling-free control and skew-aware alternatives at
31 specified counts, including an interior point of the warning range. A short synthetic external-control
32 illustration (Section 3) separates a related question: whether the reference names the intended
33 population. It supports the interpretation of the audit rather than providing a second empirical case.

34 **Relation to existing validation work.** Simulation methodology already separates estimands, meth-
35 ods and performance measures [Burton et al., 2006, Morris et al., 2019]; ICH E9(R1) specifies

the population targeted by the clinical question [International Council for Harmonisation, 2019]. Generator–analysis alignment is studied through congeniality [Meng, 1994], semi-synthetic benchmarks [Franklin et al., 2014, Curth et al., 2021], interventional evaluation [Gentzel et al., 2019], and the inverse crime [Wirgin, 2004, Kaipio and Somersalo, 2007]. Generators can also disadvantage good estimators [Shaw et al., 2026]. Simulation-based calibration checks Bayesian computation using prior-predictive ranks [Talts et al., 2018], with sensitivity depending on the test quantities [Modrák et al., 2025]. Our frequentist assessment conditions on empirical pools and specified effects; it applies established validation principles to the claim made for this artifact.

Connection to patient simulation. Digital Twin Generators learn conditional clinical trajectories [Alam et al., 2024]; ClinicalGAN evaluates personalized sequence generation and trial-dropout prediction [Chandra et al., 2024]. Those modeling tasks differ from assessing the coverage of a downstream treatment-effect interval. Our audit addresses the latter validation layer, without evaluating either system. A supporting learned-generator check shows why generator complexity does not break an estimator–reference identity (Appendix C). Clinical transfer additionally requires source comparability, as in held-out external-control validation [Ventz et al., 2019], and an explicit causal design [Hernán and Robins, 2016, Schuler et al., 2022]. These requirements define the additional evidence needed for clinical transfer.

2 Audit of a cell-count reporting rule

Application and reporting policy. The biological application separates two explanations for differentiation-marker loss in colorectal cancer: fewer marker-producing cells (a *compositional* change), or lower output from the cells that remain (an *intrinsic* change). These explanations suggest different biological interpretations. An empirical comparison of per-cell means requires the relevant population to be observed in both tissues. Let f_N, f_T denote the relevant cell fractions in reference and diseased tissue, and m_N, m_T their mean per-cell output. A reference-based decomposition, related to demographic standardization and Oaxaca–Blinder decompositions [Kitagawa, 1955, Oaxaca, 1973, Blinder, 1973], is

$$\underbrace{(f_T - f_N)m_N}_{\text{compositional}} + \underbrace{f_N(m_T - m_N)}_{\text{intrinsic}} + \underbrace{(f_T - f_N)(m_T - m_N)}_{\text{interaction}} = f_T m_T - f_N m_N. \quad (1)$$

The interaction is reported separately. If no relevant cells are observed in an arm, its sample mean is undefined; the identity cannot supply the missing mean. In a synthetic population with no such cells, the corresponding conditional mean itself has no referent. Substituting zero in either case turns an unavailable contrast into an apparent effect.

The implemented reporting rule returns an estimate at $n \geq 50$, flags a wide interval at $20 \leq n < 50$, and returns NOT ESTIMABLE below 20, where n is the relevant-cell count in diseased tissue. Unavailable reported contrasts are stored as None, not zero. The diagnostic oracle calculations examined below also retain raw arithmetic before this reporting gate. We evaluate the boundary directly and retain pre-gate intervals at smaller counts to diagnose the curve’s behavior.

The claim being re-evaluated. An internal validation statement (record A) treated the 50-cell boundary as validated because recovery ratios were 0.85–1.08 above it and 1.18 in the 20–49-cell band, using six draws per setting at 1,200 cells per arm. The design specification (record B) instead required at least 90% interval coverage and 80% exclusion of zero at a halving of expression. Appendix F and the anonymized supporting record make this mismatch inspectable.

2.1 Simulation and inferential targets

The design document fixed a detectable halving of per-cell output ($s = 0.5$), coverage of the target in at least 90% of defined outputs, and discrimination of at least 80%, defined as the probability that the interval excludes zero at that effect. The implementation scored coverage against a draw-dependent reference benchmark; we retain that score and add coverage against the generating pool mean. We call the first tested count meeting both criteria a proposed reporting cutoff, and the first meeting coverage alone a proposed warning cutoff. The implemented cutoffs are 50 and 20. Passing at one count does not require larger counts to qualify.

84 The calibration uses paired subsets of the published colorectal atlas [Lee et al., 2020]: ten patients from
 85 GSE132465 (SMC) and six from GSE144735 (KUL3). We analyze GUCA2A in the differentiated
 86 population under fixed lineage labeling. Each draw samples, with replacement, 2,000 cells per arm
 87 from an eligible two-patient pool; the reference mature fraction is $f_N = 0.40$. The implemented pool
 88 combines cells from both tissues before drawing either synthetic arm. Because the design document
 89 did not specify tissue eligibility, we also evaluate a reference-tissue-only pool. Both arms sample the
 90 same eligible pool, with an imposed expression shift. A nominal 95% percentile interval resamples
 91 mature cells in both arms, with fractions held fixed and 200 inner draws unless stated otherwise.
 92 Undefined intervals are excluded from rate denominators and reported as abstentions (Appendix B).
 93 The stored “parametric” intrinsic reference is

$$i_{\text{req}}(D_N; s) = f_N(s - 1)\bar{Y}_N, \quad (2)$$

94 where $\bar{Y}_N$ is the mature-cell mean of the *drawn* reference arm. Let P be the eligible two-patient
 95 source pool, M_P its mature-cell count, and Y_j its unshifted target-gene counts. The population
 96 parameter of this finite empirical distribution is instead

$$\mu_P = \frac{1}{M_P} \sum_{j \in P, \text{mature}} Y_j, \quad \theta_P(s) = f_N(s - 1)\mu_P. \quad (3)$$

97 We compute μ_P from all eligible source cells before either synthetic arm is sampled. Given P , the
 98 generator has $E[\bar{Y}_N] = \mu_P$ and $E[\bar{Y}_T] = s\mu_P$, making Equation 3 its true intrinsic mean contrast.
 99 It is fixed across cell draws from that pool. Patient-pair selection changes P ; reported coverage
 100 averages this conditional-pool question over the selected pairs. An all-patient pooled mean would
 101 use a different population from the generator’s two-patient draw. Both targets are conditional on the
 102 source-pool design stated above.

103 The historical grid and precision follow-up separate selection from direct evaluation; the additional
 104 repair comparison is described in Section 2.5. The controlled eight-seed grid retains inclusion of
 105 Equation 2. An initial 200-replicate diagnostic reproduced its first-seed results and added fixed-pool
 106 coverage and null rejection. The precision follow-up uses 5,000 fresh studies per cohort, pool, count
 107 (5, 50, 100, 800) and shift ($s = 0.5, 1$), split across five seed streams: 160,000 studies in total. It
 108 samples the exact marginal law of mature-cell counts in the original generator; non-mature cells
 109 do not enter this contrast. On each study we compare the original 200-resample percentile interval
 110 with a Welch–Satterthwaite interval [NIST/SEMATECH]. This unequal-variance t approximation
 111 provides an interval with no bootstrap resampling noise; it is not chosen for robustness to skewness.
 112 All 320,000 intervals are finite. The largest binomial Monte Carlo standard error is 0.71 percentage
 113 points, conditional on the empirical source pools. Appendix A specifies the sampling and interval
 114 calculations.

115 2.2 Fixed-pool coverage and false positives

116 Two standards answer different questions (Table 1). At $s = 0.5$, percentile coverage is below the
 117 historical minimum of 90% in three conditions; SMC/reference reaches 91.0% [90.2, 91.8]. All four,
 118 including SMC/reference, remain below the intervals’ nominal 95% level, with every Wilson upper
 119 endpoint below 95%. The historical tolerance does not establish nominal calibration.

120 At $s = 1$, the fixed-pool target is zero, so null rejection is exactly one minus coverage among
 121 defined intervals. The reported 9.8–32.2% null rejection is therefore 67.8–90.2% null coverage,
 122 rather than an independent calibration criterion. Coverage is lower at the null than at the halving in
 123 all four conditions. Detection at $s = 0.5$ is a distinct quantity: exclusion of zero under a nonzero
 124 effect, here 47.5–76.2%. All detection upper endpoints are below the historical 80% requirement; all
 125 null-rejection lower endpoints exceed nominal 5%. Wilson intervals quantify Monte Carlo precision
 126 conditional on the source pools.

127 2.3 Shared-reference scoring removes part of the estimation error

128 At 800 cells, evaluating the percentile intervals against the sampled reference gives 98.9–99.6%
 129 inclusion, compared with 92.3–93.6% coverage of the fixed-pool effect. The paired differences are
 130 5.8–6.8 percentage points with Monte Carlo standard errors of 0.33–0.36 points. Only the scoring

Table 1: At the 50-cell reporting boundary, substituting Welch intervals improves coverage but does not restore the required performance. Rates (%) and 95% Wilson Monte Carlo intervals use 5,000 studies per cohort, pool and effect. Methods share the same samples. Coverage targets the fixed-pool effect at $s = 0.5$; null rejection uses $s = 1$; detection uses $s = 0.5$. Historical requirements are at least 90% alternative coverage and 80% detection. Nominal coverage is 95%; null rejection is one minus null coverage and should be 5%.

Cohort / pool	Interval	Coverage	Null rejection	Detection
SMC / pooled	Percentile	87.1 [86.2, 88.0]	25.4 [24.2, 26.6]	51.1 [49.7, 52.4]
SMC / pooled	Welch	90.2 [89.3, 91.0]	23.5 [22.3, 24.7]	50.7 [49.3, 52.1]
SMC / reference	Percentile	91.0 [90.2, 91.8]	9.8 [9.0, 10.7]	76.2 [75.0, 77.3]
SMC / reference	Welch	92.5 [91.8, 93.2]	8.9 [8.1, 9.7]	77.6 [76.5, 78.8]
KUL3 / pooled	Percentile	84.7 [83.6, 85.6]	32.2 [31.0, 33.5]	49.0 [47.6, 50.4]
KUL3 / pooled	Welch	89.7 [88.8, 90.5]	30.8 [29.5, 32.1]	49.0 [47.6, 50.3]
KUL3 / reference	Percentile	86.3 [85.3, 87.2]	19.0 [18.0, 20.1]	47.5 [46.1, 48.8]
KUL3 / reference	Welch	88.1 [87.2, 89.0]	18.8 [17.7, 19.9]	48.2 [46.8, 49.6]

target changes. At five cells the direction reverses; shared randomness need not inflate inclusion at every sample size.

The increasing discrepancy at large tumor-arm counts has an analytic explanation. The estimate is $\hat{\theta} = f_N(\bar{Y}_T - \bar{Y}_N)$, whereas its error against the drawn benchmark is $\hat{\theta} - i_{\text{req}} = f_N(\bar{Y}_T - s\bar{Y}_N)$. For independent arm draws conditional on P , their error variances differ by

$$\text{Var}(\hat{\theta} - \theta_P | P) - \text{Var}(\hat{\theta} - i_{\text{req}} | P) = f_N^2(1 - s^2) \frac{\sigma_P^2}{n_N}, \quad (4)$$

where σ_P^2 is the source-pool expression variance and $n_N = 800$. At $s = 0.5$, the drawn benchmark removes three quarters of the reference-arm variance contribution from the error being assessed, while the bootstrap continues resampling both arms. This explains why near-perfect benchmark inclusion can accompany lower coverage of the generating parameter. Skewed finite-sample errors still determine the actual percentile-interval rates.

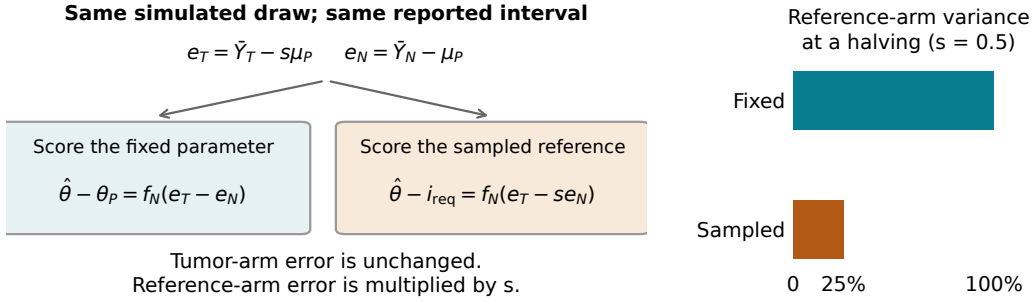


Figure 1: How scoring changes the error while keeping the reported interval fixed. The schematic uses centered arm means conditional on pool P . At a halving, the sampled-reference score halves the reference-arm error, reducing that arm's variance contribution to one quarter. The tumor-arm contribution is unchanged; the bars show reference-arm variance only.

The historical grid retains its original sampled-reference score; it does not supply fixed-pool calibration.

2.4 Welch as a control for bootstrap resampling noise

Welch intervals increase fixed-pool coverage at 50 cells by 1.5–5.0 percentage points on the same draws (paired Monte Carlo standard errors 0.23–0.32 points). They also reduce null rejection, but it remains 8.9–30.8% and every lower confidence endpoint exceeds 5%. Welch intervals have no bootstrap resampling noise, so their excess null rejection on the same draws shows that finite inner-bootstrap noise cannot be the sole explanation for the failure shared by both methods. Detection

remains below 80% even at the upper confidence endpoints. Substituting Welch intervals therefore fails the joint requirements at 50 cells. Section 2.5 separately compares bootstrap budgets and skew-aware methods.

Why detection can fall as cell count rises. The apparent detection curve combines effect sensitivity with a changing false-positive rate. Figure 2 reproduces the pooled-draw trough with greater precision. For KUL3, percentile detection falls from 49.0% at 50 cells to 44.0% at 100, then rises to 57.7% at 800; null rejection falls from 32.2% to 20.3% to 6.5%. At 50 and 100 cells, the median sample-based standard error is respectively 0.66 and 0.86 times the exact pool-conditional standard deviation. The fraction of draws with that ratio below one half falls from 29.1% to 7.6%. These diagnostics support a loss of spuriously confident rejections as the sampled variance becomes better represented, before increasing signal-to-noise dominates. All-zero tumor samples occur in only 0.32% of the 50-cell draws and none at 100: the trough is not explained by all-zero samples alone.

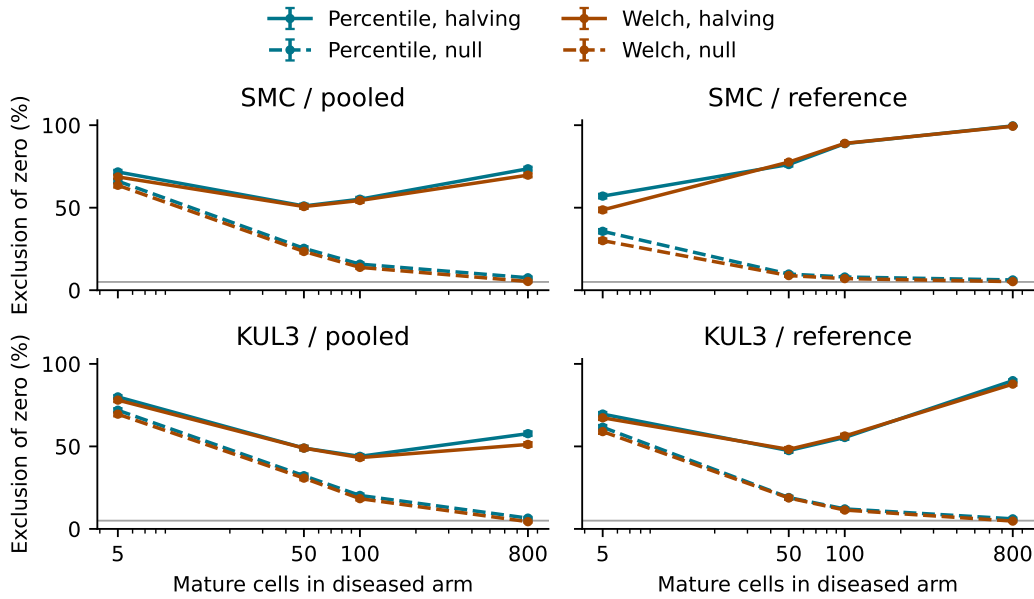


Figure 2: Detection must be read alongside null rejection. All four cohort/pool conditions use 5,000 studies per point; bars are 95% Wilson Monte Carlo intervals. Solid curves use a halving of expression and dashed curves the null. The horizontal line marks nominal 5% rejection. Five-cell intervals are diagnostic calculations before the reporting gate, which abstains below 20. Both methods use the same samples; their apparent sensitivity at sparse counts accompanies excessive false positives.

The five-cell diagnostic makes the problem especially clear: pooled percentile null rejection is 66.1% in SMC and 71.9% in KUL3. Under the halving, 46.2% and 41.9% of tumor samples are all zero. Resampling cannot recover an unobserved positive tail, although the reference arm still contributes uncertainty. The original reporting gate already abstains here; the 50-cell failures concern returned answers directly.

2.5 Testing the warning range and alternative intervals

A separate fixed-design follow-up evaluates 30 and 50 cells, using 2,000 fresh studies per cohort/pool/shift: 32,000 studies and 160,000 interval evaluations. It compares percentile intervals with 200 and 2,000 resamples, bias-corrected and accelerated (BCa) and bootstrap- t intervals with 2,000 resamples, and Welch. These skew-aware methods follow established bootstrap accuracy work [DiCiccio and Efron, 1996]. All methods use the same study samples; bootstrap methods share resamples. Coverage uses the fixed-pool target. These fresh draws differ from Table 1, explaining small differences at the shared 50-cell setting. Appendix A specifies implementation and numerical checks.

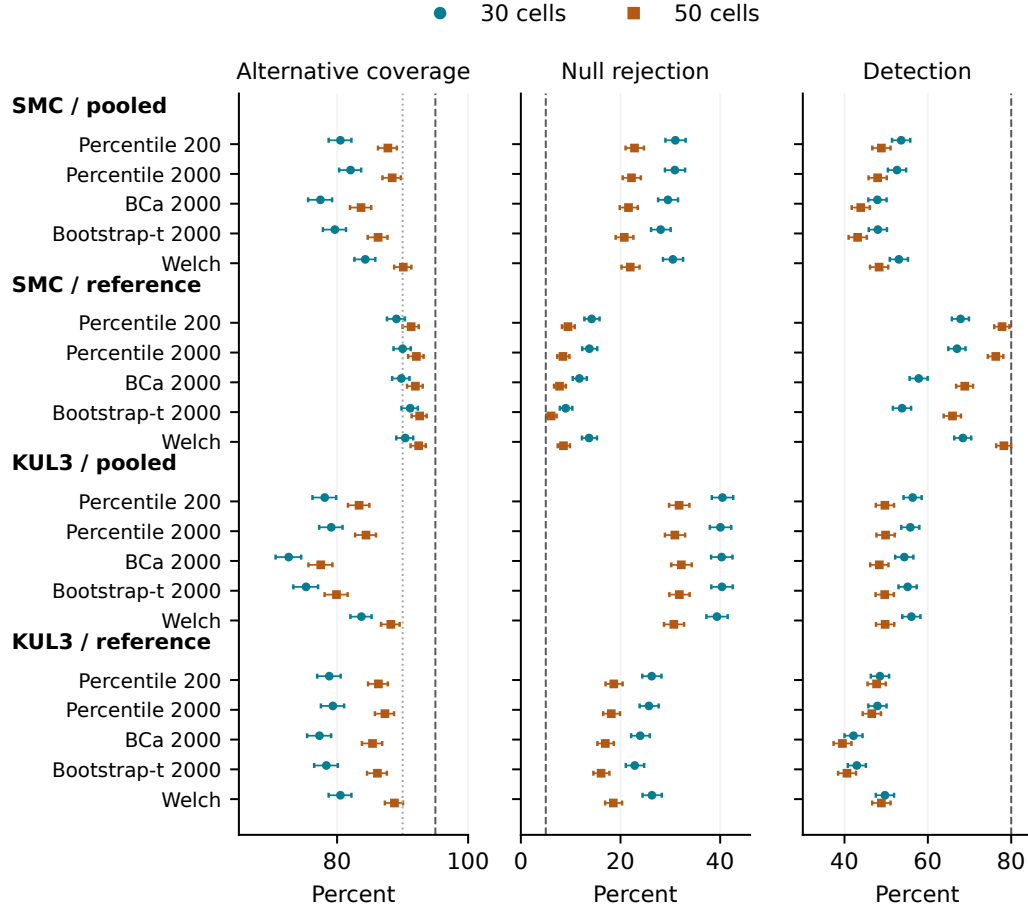


Figure 3: All five intervals at 30 and 50 cells, with pointwise 95% Wilson Monte Carlo intervals (2,000 studies per condition/effect). Method suffixes are bootstrap budgets. Alternative coverage and detection use the halving; null rejection uses no shift. Dashed lines mark nominal 95% coverage, 5% null rejection and required 80% detection; the dotted line marks the historical 90% coverage minimum. All intervals are defined. Exact rates and paired comparisons accompany the figure.

At 30 cells, the original interval has 78.1–89.1% alternative coverage, 14.2–40.5% null rejection and 48.5–67.9% detection. Flagging its width does not establish calibration. This evaluates one interior point, not the whole 20–49-cell band. At 50 cells, increasing the percentile budget reduces null rejection by 0.5–1.0 percentage points (paired Monte Carlo SEs 0.28–0.34 points), much less than its excess over nominal.

The strongest null improvement occurs for bootstrap- t in SMC/reference: rejection falls from 8.45% for percentile with 2,000 resamples to 6.05% [5.09, 7.18], a paired reduction of 2.40 points (SE 0.40). Alternative coverage is 92.6%, but detection falls from 76.3% to 65.9%. The improvement does not transfer uniformly: in KUL3/pooled, BCa and bootstrap- t have 32.2% and 31.8% null rejection, against 30.9% for the equal-budget percentile interval. All methods return intervals, so these comparisons are not driven by abstention. No tested method meets the historical coverage/detection requirements together with nominal null calibration at either count. This bounds the tested repair attempt without selecting a new rule.

Patient resampling: a different target, not an automatic repair. The project requires patients as the unit for patient-population inference. Our cell-draw audit instead evaluates uncertainty conditional on a fixed source pool. To test the proposed patient-level repair directly, we rescore the paired patient- t interval on the original simulated draws against both fixed cell-weighted and equal-patient targets (Appendix A). At 50 cells and two held-out patients, fixed cell-weighted coverage is

193 96.0–97.8%, while detection is only 4.2–8.0% and alternative abstention is 0.35–34.55%. Thus the
 194 coverage/detection tradeoff survives correcting the scoring target. But increasing the holdout to five
 195 patients leaves KUL3 null rejection at 8.7–18.1% at 50 cells. Neither near-nominal rejection in some
 196 settings nor wider intervals establishes a general repair. These studies use two or five source patients
 197 per draw, not the full cohorts of ten and six; they cannot establish the power of a patient-population
 198 study of those cohorts.

199 **Source shape and arm balance.** A separate sensitivity study uses 2,000 studies and 2,000 percentile
 200 resamples per setting (Appendix A). At 50 tumor cells, empirical-pool null rejection is 9.70–31.80%;
 201 removing zeros and rescaling the positive counts to the original mean gives 7.10–12.65%, while
 202 a mean/variance-matched Gaussian gives 5.45%. Keeping the empirical pools but changing the
 203 reference count from 800 to 50 gives 7.45–11.45%. This implicates source shape and arm imbalance
 204 without isolating their individual contributions. Under the null, equal-sized independent arms have
 205 an exactly symmetric difference by exchangeability, even when the source is skewed. Balancing
 206 also changes variance and the reference sample size; alternative detection was not evaluated for this
 207 design change, so these rates do not justify discarding reference cells or establish a repair.

208 2.6 A first passing count depends on the evaluated grid

209 A grid is the set of counts tested in a simulation. We compare the original set
 210 $\{0, 20, 40, 100, 200, 400, 800\}$ with two expanded sets, using the same simulated outputs at shared
 211 counts (eight streams; 200 studies and 200 bootstrap draws per setting). With fixed bin boundaries,
 212 SMC’s proposed reporting cutoff moves from 100 to 70 in all eight streams. Yet the first passing
 213 individual counts span 50–80 and 45–80 on the two expanded grids. A binned cutoff is the median
 214 count in a qualifying bin, not a measured crossing location. KUL3 returns 400 in six streams and
 215 800 in two; pooled draws yield no qualifying cutoff. These exploratory rates have Monte Carlo SEs
 216 up to 3.5 percentage points. Unresolved local crossings at counts 45, 60 and 70 cannot establish a
 217 nonmonotone performance curve. Appendix B and the saved grid tables retain the settings.

218 2.7 Patient influence and the cohort gap

219 Enumerating all 45 and 15 two-patient holdout pairs separates patient omission from repeated
 220 simulation. Keying streams to the sorted pair identifiers keeps every retained pair fixed. Raising
 221 the inner bootstrap budget from 200 to 1,000 and 5,000 leaves the first cohort’s proposed cutoff at
 222 70, but moves the second’s 400 to 300 and back to 400. Omitting one source patient changes the
 223 joint joint cutoff for none of ten patients in the first cohort and three of six in the second (KUL01,
 224 KUL30 and KUL31). For the coverage-only cutoff, the corresponding counts are five of ten and zero
 225 of six. These checks quantify influence within each observed cohort. Appendix B gives the biological
 226 comparability limits.

227 The sensitivity windows cover counts 40, 50, 60, 70, 80, 90, 100, 110 in SMC and
 228 200, 300, 400, 600, 800 in KUL3. These per-count summaries use a fixed pair schedule, so
 229 they are a separate experiment from the eight-seed union grid. Omitting KUL01 or KUL30 moves
 230 the joint cutoff from 400 to 300; omitting KUL31 moves it to the lowest tested count, 200, with its
 231 lower boundary unobserved. The coverage-only baselines likewise occur at the lowest tested counts,
 232 40 and 200. Their stability cannot locate a lower crossing outside those windows. Overlapping pairs
 233 do not add independent patients.

234 **Expression dispersion partly explains the cohort gap.** Holding tissue eligibility fixed at reference-
 235 only, the complete SMC and KUL3 mature-cell pools contain 662 and 844 cells. Across their 45 and
 236 15 eligible patient pairs, median source-pool squared coefficients of variation ($c_P^2 = \sigma_P^2 / \mu_P^2$) are 4.35
 237 and 16.99; median zero-expression fractions are 39.3% and 62.1%. The source means are 19.5 and
 238 11.9 raw counts, respectively (medians over pools). The KUL3 contrast therefore has substantially
 239 more relative sampling noise despite its larger source-cell pool.

240 For the implemented binomial thinning, the tumor-cell variance is $s^2 \sigma_P^2 + s(1-s)\mu_P$. At $s = 0.5$,
 241 the exact conditional variance of the contrast relative to its squared signal is

$$\frac{\text{Var}(\hat{\theta} | P)}{\theta_P(0.5)^2} = \frac{c_P^2 + 1/\mu_P}{n_T} + \frac{4c_P^2}{n_N}. \quad (5)$$

With $n_T = 50$ and $n_N = 800$, plug-in signal-to-standard-deviation ratios using the median pool moments are 3.02 in SMC and 1.53 in KUL3. This provides a distributional explanation for weaker KUL3 discrimination. It does not reproduce the exact 70-versus-400 cutoff ratio: percentile-bootstrap behavior, pair heterogeneity and binning also affect that selection. The biological causes of the source-distribution difference remain unexplained.

Biological interpretation is a separate requirement. Interval calibration cannot establish a biological mechanism. The assumed retained-gene control, MS4A12, instead falls by -0.971 to -0.951 in non-sparse primary-cohort strata. Its depletion persists under broader cell inclusion and depth matching and appears in a separate bulk dataset. MS4A12 therefore does not serve as a retained-gene control in these data, a distinct failure of the panel assumption. These observations do not separate composition from within-cell change. Appendix E reports this control-panel audit as a limitation of the biological application.

3 Population alignment in clinical simulation

A clinical simulation must also name the population whose effect is evaluated. In a constructed external-control example, the intended target τ_E is the average individual treatment effect among enrolled records. Averaging stratum contrasts over the pooled trial-plus-control population instead estimates a different target. Scoring that estimator against the same pooled contrast gives perfect agreement by construction. More informatively, a pooled potential-outcome reference breaks the identity yet still favors the wrong-population estimator; scoring against τ_E reverses the ranking. Under zero case-mix shift the mismatch nearly disappears. This separates target alignment from numerical recovery; Appendix D gives the simulation, controls and results.

4 Implications for clinical-simulation validation

The audit measures a failure of the recorded reporting claim; the synthetic illustration explains why target-population alignment is a separate requirement. Three established checks make these distinctions operational.

State the target before constructing its reference. Distinguish a population treatment effect, a latent finite-cohort effect and an observed-data contrast. Name the population and the averaging weights. A reference need not be independent of the simulated dataset: τ_E is necessarily a property of the enrolled records. The requirement is that it represents the intended target, rather than merely reproducing the estimator.

Evaluate the interval and reporting policy jointly. Assess the whole decision rule using the interval that would actually be reported. Record bias and error against the chosen target, null rejection, detection at the specified effect, and abstention frequency. Report both conditional interval performance among returned answers and how often a valid answer is returned. A successful effect-detection rate can accompany excessive null rejection, so the former alone cannot validate an interval. For a nominal 95% interval, null rejection and null noncoverage are the same quantity; alternative coverage and detection answer different questions.

Separate rule selection from assessment and transfer. Separate selecting a rule from assessing it. If a scalar threshold is intended to guarantee performance above that threshold, qualification must address the evaluated upper range rather than only its first passing point. Reserve independent simulation draws for the final assessment, report rate uncertainty and the tested range, and distinguish this from transfer to new patients or cohorts. Increasing resamples from an empirical pool improves conditional numerical precision; it does not broaden that pool's biological support. For an external-control application, the additional question is comparability across data sources, which requires direct study-level validation such as the held-out comparisons of Ventz et al. [2019].

5 Conclusion

The audit separates three claims that had been conflated: recovering a reference value, covering the intended parameter, and meeting the requirements of a reporting rule. Sharing the sampled reference mean changes the error variance being assessed. Direct evaluation of the fixed-pool target finds undercoverage at the halving and still worse coverage at the null. Increasing the bootstrap budget and using skew-aware intervals improve some conditions but do not qualify either tested reporting count.

For this artifact, the practical action is to withdraw the claim that 50 cells is a validated reporting boundary. Outputs can be retained as exploratory estimates with unvalidated uncertainty; a warning about width is not a calibration guarantee. A replacement needs an explicit acceptable error rate, adequate detection and independent assessment across its intended reporting range. New patient-level evidence is needed before extending these conditional results to clinical decisions.

6 Responsible use, limitations and impact

Misuse and limitations. NOT ESTIMABLE must not be read as “no effect” or as evidence that surviving cells are behaving normally. A separate within-cell analysis over three cohorts and 42 patients also remained unresolved because its comparability checks returned *undecided* (Appendix B). The central evidence is a data-based simulation self-audit; neither it nor the constructed examples validate a patient world model or a replacement reporting rule. The scope and conditional target are stated in Section 1. Repeated cell draws and overlapping patient pairs do not measure uncertainty over new patients. Poisson- and negative-binomial-based intervals remain untested; one point in the warning range does not qualify its full range. The draw-pool choice was originally underspecified. The absence-defined labeling axis remains sensitive to sequencing depth (Appendix B). The external search found no affirmative case, so the documented failure remains a self-audit rather than evidence about its prevalence. The external-control example uses a synthetic registry and an illustrative decision threshold; it does not establish clinical benefit or support a treatment decision.

Data and impact. The single-cell cohorts are public, de-identified and previously published. The release policy covers derived summaries and code, rather than cell-level matrices. A system reporting group-specific quantities should distinguish an unavailable comparison from an estimated null and should evaluate the coverage and discrimination of any abstention rule. Publishing failed calibration attempts helps prevent unsupported thresholds from being treated as validated.

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412 A Simulation settings

Experiment	Fixed settings and sampling units
Single-cell calibration	GSE132465 (SMC): 10 paired patients; GSE144735 (KUL3): 6. Target: GUCA2A; <code>stem_pole</code> axis, <code>lineage</code> resolution, <code>differentiated</code> mature label; reference (<code>normal</code>) weighting. Each replicate holds out two patients and draws 2,000 cells per synthetic arm from their eligible pool; reference mature fraction 0.40. Expression shifts are 1.0, 0.8, 0.5, 0.25; calibration uses 0.5.
Interval construction	A nominal 95% percentile interval from 200 resamples of mature cells within each synthetic arm, holding cell fractions fixed. This assesses the conditional intrinsic contrast; it does not estimate population uncertainty across patients. Coverage and discrimination condition on defined intervals. The historical inclusion score uses the drawn reference in Equation 2; the paired follow-up also evaluates fixed-pool coverage using Equation 3.

414 **Precision follow-up and comparator.** Five fresh seed streams (202609160–202609164), each
415 with 1,000 studies per setting, provide 5,000 draws at each of four counts and two shifts in four
416 cohort/pool combinations. Every draw uniformly chooses a two-patient pair, samples 800 reference
417 and n_T tumor mature-cell counts independently with replacement, and thins tumor counts binomially.
418 This reproduces the generator’s marginal law; all 120 source-pair summaries match. Percentile and
419 Welch implementations agree with independent checks.

420 Writing $v_a = S_a^2/n_a$ for arm a , the Welch interval is $\hat{\theta} \pm t_{0.975,\nu} f_N \sqrt{v_N + v_T}$, where $\nu =$
421 $(v_N + v_T)^2 / [v_N^2/(n_N - 1) + v_T^2/(n_T - 1)]$. Zero estimated total variance would give an un-
422 available comparator interval; none occurred. For the sparsity diagnostic, sample-based standard
423 error is compared with $f_N \sqrt{\sigma_P^2/n_N + (s^2\sigma_P^2 + s(1-s)\mu_P)/n_T}$. Settings were fixed before this
424 exploratory follow-up; aggregate exports retain widths, seed rates and paired differences.

425 **Reporting-band and repair follow-up.** Five fresh streams (202609170–202609174), 400 studies
426 each, use the same pool-conditional sampling law at 30 and 50 cells. For BCa, bias correction uses
427 midpoint ties; acceleration uses signed two-sample influence contributions $-(Y_{Ni} - \bar{Y}_N)/n_N$ and
428 $(Y_{Ti} - \bar{Y}_T)/n_T$. Endpoints agree with SciPy’s independent two-sample jackknife implementation
429 on continuous, discrete, sparse and zero-tumor samples. Bootstrap- t uses $t^* = (\hat{\theta}^* - \hat{\theta})/\widehat{SE}^*$ and
430 reversed pivot quantiles, with Welch standard errors recomputed in each resample. Invalid BCa
431 transforms or any undefined studentized pivot return an unavailable interval; none occurred. All
432 160,000 intervals are finite. The study settings were fixed before the run; this remains exploratory
433 analysis on existing pools.

434 At least one adjusted BCa endpoint lies within the outermost ten bootstrap order positions in 5.2–
435 56.3% of studies across settings. These finite-budget results therefore do not characterize BCa with
436 arbitrarily precise tail quantiles. Seed rates, paired differences and numerical diagnostics accompany
437 the aggregate summaries.

438 **Source substitutions and clustered alternatives.** The separately registered diagnosis crosses
439 two counts (50, 800), four cohort/pool conditions, four source families and two arm-size designs.
440 Each of its 64 rows uses five streams of 400 draws, 2,000 inner resamples and the same two-patient
441 pools; all intervals are defined. The families are empirical counts, mean-rescaled positive counts,
442 mean/skew-matched gamma (with some skew clipping), and mean/variance-matched Gaussian. All
443 rates were reproduced from source data. Gaussian rows reuse standardized draws across cohorts.
444 Substitution contrasts are not additive causal shares or lower bounds on a skew contribution. At 800
445 cells the designs coincide; empirical null rejection is 4.55–6.40%.

446 **Patient- t target check.** We replay the incoming patient- t studies at 50 and 800 cells, two and five
447 held-out patients, both shifts and all four cohort/pool conditions: 64,000 studies (five streams of
448 400 per setting). Returned intervals, rejection, abstention and sampled-reference inclusion counts
449 match all 32 original settings exactly. The two fixed targets use either the mean of all eligible source
450 cells or the equally weighted mean of source-patient means, computed before drawing arms. Both
451 condition on the selected pool. Patients lacking mature source cells are outside the latter mean;
452 incomplete patient pairs can make the interval unavailable. At 50 cells and two held-out patients,

equal-patient coverage is 96.1–97.6%. Exports include all counts, target-specific coverage, Wilson intervals, availability and unconditional detection.

B Calibration details and limits

Historical grids. Earlier reference-only SMC cutoffs were 100, 90 and 70 under grids with 500, 500 and 200 replicates. They changed streams, bins and budgets together; Section 2 uses shared draws to separate grid effects. KUL3’s largest setting resampled 800 mature cells per arm from 844 distinct source cells, not additional patients.

Recovery accounting and unavailable contrasts. Across 1,711,200 diagnostic replicates, 427,800 have zero requested reference and undefined recovery ratios. Estimate–realized-reference residuals are zero. At zero cells, raw diagnostic values arise from setting an empty-arm mean to zero; they define no effect in absent cells. The reporting gate abstains. Corrected interval rates exclude undefined outputs and report availability; an all-abstaining bin has no coverage estimate. The saved summaries retain the full sweep accounting.

Biological scope and patient uncertainty. Across three cohorts and 42 patients, patient-bootstrap expression intervals cross the accepted $\log_2$ boundaries $[-0.5, 0.5]$, leaving comparability undecided. Depth matching discards about three quarters of cells. Changing SMC tissue eligibility changes the mature-cell pool from 662 to 3,824; Section 2.7 holds eligibility fixed.

C Supporting estimator and trial checks

Reference dependence. With nonempty stratum arms, saturated G-computation [Robins, 1986], inverse-probability weighting (IPW) using empirical stratum propensities, non-cross-fitted saturated augmented IPW (AIPW) and matching using every control all reproduce $T(D) = \sum_g (n_g/n)(\bar{Y}_{1g} - \bar{Y}_{0g})$. The IPW weights recover each stratum mean; the AIPW augmentation sums to zero within stratum. Ordinary least squares (OLS) with additive stratum indicators instead weights contrasts by normalized $n_g \hat{p}_g(1 - \hat{p}_g)$ [Angrist, 1998, Frisch and Waugh, 1933]. These known identities show why the reference determines what an exact recovery check establishes. Exact 1:1 allocation within strata makes OLS, standardized and unadjusted contrasts coincide; balance only in expectation does not guarantee this identity.

Learned-generator check. Four Gaussian mixtures, a multilayer-perceptron conditional Gaussian and two parametric controls generate synthetic cohorts after fitting to simulated three-stratum training data. Across 42 generator–size settings, saturated G-computation has zero residual against $T(D)$ on every evaluable dataset; this follows from the identity above. Nonsaturated fits can depart, with a largest MLP residual of 3.3 outcome units under mixture generators. The aggregate archive retains all scheduled/evaluable counts and exclusions. Generator complexity alone does not make an identity-based check informative.

D Synthetic external-control design

Table 2: Reference choice reverses the estimator ranking in the synthetic external-control example. Bias and RMSE use 200 studies with 1,600 enrolled records and 4,800 controls. Columns identify the population used to average observed stratum contrasts; only enrolled standardization matches the intended target. All entries are in abstract outcome units.

Validation reference	Pooled standardization		Enrolled standardization	
	Bias	RMSE	Bias	RMSE
Observed contrast, pooled	0.000	0.000	-0.872	0.876
Potential outcomes, pooled	+0.003	0.115	-0.869	0.875
Potential outcomes, enrolled	+0.885	0.892	+0.013	0.105

For enrolled records E and controls C , the target is $\tau_E = |E|^{-1} \sum_{i \in E} \{Y_i(1) - Y_i(0)\}$. The estimator averages stratum contrasts using either enrolled shares $n_{Eg}/|E|$ or pooled shares $(n_{Eg} + n_{Cg})/(|E| + |C|)$. At zero case-mix shift, pooled-standardization bias against τ_E is -0.002 units. Under the full shift, between-arm balance coexists with a maximum standardized difference of 0.964 from the enrolled population.

The registry has 50,000 records with stratum shares $(0.55, 0.30, 0.15)$, $Y_0 \mid g \sim \mathcal{N}(\mu_g, 3^2)$, $\mu = (12, 8, 4)$, and $Y_1 - Y_0 = \delta_g r$, $\delta = (3, 1.5, 0.25)$, $r \sim \text{Gamma}(4, 0.25)$ (shape, scale), hidden from estimators. Enrollment favors $(0.15, 0.25, 0.60)$; disjoint arms sample without replacement. These are abstract continuous outcomes. The pooled population combines trial and control records.

One seed (20260915) generates the registry and 200 studies at each of $n = 50, 100, 200, 400, 800, 1600$. Eight estimators and five references give 48,000 comparisons: 47,969 finite, with 31 exclusions from one empty-stratum study at $n = 50$. Five case-mix shifts and 50 balance draws at $n = 1600$ provide controls. Saved threshold crossings compare medians, not individual decisions. All nine prespecified falsification checks pass.

E The retained-gene control

The control-panel audit compares saved unmatched and depth-matched single-cell summaries with a separate bulk dataset. Where both mature fractions equal one (all epithelium, without marker-based subdivision), MS4A12 still changes by -0.967 to -1.000 across the tested cohort/matching strata; CDX2 on the same cells changes by -0.396 to -0.293 . Depth-matched MS4A12 changes are -0.971 to -0.960 . These observations weaken a uniform labeling or depth artifact explanation; they do not eliminate every measurement artifact. Across nondegenerate strata, the median baseline is 2.56 CP10K and tumor expression 0.104 CP10K. Strata classified as degenerate have a median of one relevant tumor cell and cannot establish a precise complete loss.

In GSE39582 (585 bulk samples), the saved MS4A12 $\log_2$ fold change is -8.18 . This corroborates tissue-level depletion on a separate platform; it does not identify a within-cell mechanism. Lower-abundance genes retain more signal in the single-cell summaries, arguing against a simple universal abundance-floor explanation, not proving absence of gene-specific dropout. Repository chronology places panel freezing before dataset registration; it cannot establish what evidence was known outside the repository. The panel's prior expectation of retention therefore needs reassessment.

F External audit and research context

Search and classification. Of 41 queued entries, 26 were examined: 19 screened out and seven classified. A pre-inspection amendment relaxed unsuccessful queries without changing the rule. All verdicts were NO: six use generating/outside-sample references; the seventh's near-zero residual on noiseless data becomes 0.125 with noise. One reader classified this nonrandom sample and executed only the seventh repository. No examinable virtual-control-arm code was found. The search supports no prevalence estimate.

Related scope. Cell-count filters [Crowell et al., 2020], composition analysis (scCODA, Milo, propeller and cacao) [Büttner et al., 2021, Dann et al., 2022, Phipson et al., 2022, Petukhov et al., 2022], selective prediction [Chow, 1970, El-Yaniv and Wiener, 2010, Geifman and El-Yaniv, 2017] and conformal coverage [Vovk et al., 2005, Angelopoulos and Bates, 2023] address related tasks; none automatically validates this intrinsic-effect interval or its reporting rule.

Documentary and implementation provenance. The anonymous record includes the original validation statement (Artifact A), interval criteria (Artifact B), and numerical context. Exact source mappings remain outside the review bundle. Saved tables carry commit hashes and seeds; these establish traceability, not statistical validity.

Use of large language models. Language-model assistants supported code, figures, drafting and audits, including identifying and introducing defects. Authors remain responsible for every claim; automated checks complement their review.